Code No. 3523

OSMANIA UNIVERSITY
FACULTY OF ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (AUTONOMOUS)
B.E. (AI&ML) V-Semester (MAIN) Examinations Dec 2023 /Jan 2024

DESIGN AND ANALYSIS OF ALGORITHMS

Time : 3 hours **Max. Marks : 70**

Note : *i) Answer **Question No. 1** (Compulsory) and answer any **four questions** from the remaining questions (2- 7).*
ii) Answers must be written in same order as they occur in the Question Paper.
*iii) **Missing data, if any may suitably be assumed.***

		Marks	BT	CO
1. a)	Discuss the general plan of divide and conquer technique.	2	2	1
b)	What is the difference between feasible solution and optimal solution. Give an example.	2	1	1
c)	Compare the greedy technique with dynamic programming.	2	2	2
d)	State how Prim's algorithm and Kruskal's algorithm differ from each other.	2	1	2
e)	Contrast the Graph coloring problem with an example.	2	2	1
f)	Define general abstraction of branch and bound.	2	1	2
g)	Define state space tree.	2	1	2
2. a)	Apply merge sort to sort the list E, X, A, M, P, L, E in alphabetical order. Solve the recurrence relation for the number of key comparisons made by merge sort in the worst case.	7	3	2
b)	Calculate step count for an algorithm to perform matrix addition.	7	3	2
3. a)	Explain about the control abstraction of divide and conquer with the help of an example.	7	3	2
b)	Show the intermediate steps when numbers are sorted using quick sort: 65, 70, 75, 80, 85, 60, 55, 50, 45.	7	3	2
4. a)	Design a dynamic programming algorithm for finding an optimal order of multiplying 'n' matrices.	7	3	3

(P.T.O.)

	b)	Apply the bottom-up dynamic programming algorithm to the following instance of the knapsack problem: n=2, capacity w=6, weight = {3,2,1,4,5}, value = {$25,$20,$15,$40,$50}.	7	4	3
5.	a)	Apply backtracking algorithm write pseudocode for n-queens problem.	7	3	3
	b)	Apply backtracking to solve the following instance of the subset-sum problem: S={3, 5, 6, 7} and d =15.	7	3	2
6.	a)	Discuss about the strategy to show that the problem in NP hard.	7	2	3
	b)	Explain the general method for branch and bound. Explain how it differs from backtracking.	7	2	3
7.	a)	What are the different ways of searching an answer node in a state space tree. Explain with example.	7	2	3
	b)	Compare the relationship between NP hard and NP complete Problems with an example.	7	3	3

****